Do Semantic Distance Tests *Actually* Predict Creativity in LLMs?

Anonymous Authors[1]

Abstract

Automated semantic distance tests are increasingly used to measure the "creativity" of large language models (LLMs). However, the validity of semantic distance tests as predictors of *machine* creativity has not yet been established, and these tests already have limited validity as predictors of human creativity as it stands. To address this problem, we conduct the first systematic study evaluating the effectiveness of semantic distance tests in predicting the creative achievement of LLMs across three constructs: creative writing, divergent thinking, and scientific ideation. We score each test on two psychometric criteria (*validity* and *specificity*), and derive an upper bound on the maximum attainable specificity a test can achieve as a function of its validity and its construct's correlation with general capabilities, which we refer to as the *validity-specificity frontier*. We find that: **(1)** Test effectiveness varies significantly by construct, and no single test predicts all constructs well. **(2)** None of the tests is a good predictor of scientific ideation ability. **(3)** Existing tests are far from the validity-specificity frontier, indicating meaningful room for the design of improved tests moving forward. Our findings provide clear practical takeaways and directions for future work, identifying tests which are best suited to predict each construct, and suggesting that novel tests are needed to reliably predict scientific ideation ability.

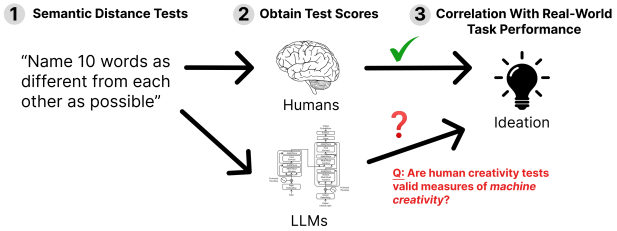

Figure 1. **Overview.** Semantic distance tests like the Divergent Association Task ask respondents to name a set of maximally dissimilar words and score the result by embedding-space distance. The external validity of these tests among humans has been supported by modest correlations with ideation tasks, but their validity as measures of *machine* creativity has not been adequately established, motivating the central question of this work: *Do semantic distance tests actually predict creativity in LLMs?*

1. Introduction

Evaluating the creativity of large language models (LLMs) is essential for developing methods that improve creativity, advancing our scientific understanding of this ability, and ensuring robust deployment of AI in human co-creativity environments. In recent years, it has become common practice to re-purpose psychological assessments of *human* creativity to purportedly evaluate the "creativity" of LLMs (Stevenson et al., 2022; Chen & Ding, 2023; Bellemare-Pepin et al., 2024). In particular, the ability to associate semantically distant concepts has long been considered central to human creativity (Mednick, 1962; Simonton, 2023), motivating the use of *automated* semantic distance assessments like the Divergent Association Task (DAT) (Olson et al., 2021; Chen & Ding, 2023; Cropley, 2023; Bellemare-Pepin et al., 2024; Wang et al., 2025) to assess whether LLMs are capable of making distant associations. On the DAT, subjects are instructed to generate ten maximally dissimilar nouns, and scores are given by the mean pairwise semantic distance of all word pairs under an embedding model such as GloVe (Pennington et al., 2014), providing a convenient way to assess "creativity" without requiring human raters.

Over the last year, two novel semantic distance tests have also been proposed. The Conditional DAT (CDAT; Nakajima et al. (2026)) extends the DAT by requiring dissimilar nouns to each be relevant to a "cue" word, incorporating a measure of appropriateness in addition to novelty (Boden, 2004; Maher, 2010). Similarly, the Parallel Association Chain Evaluation (PACE; Qiu & Hu (2025)) is inspired by the forward-flow measure (Gray et al., 2019; Beaty et al., 2021) and instructs models to make free associations starting from several seed words, scoring responses by a sequential semantic distance measure. The PACE test shows strong

[1]Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

Spearman rank correlations with creative writing benchmarks ($\rho \approx 0.74$). However, it also correlates strongly with general model capabilities ($\rho \approx 0.66$), so it is unclear how well PACE measures creative achievement *independent* of what general capability already predicts. More broadly, the validity of semantic distance tests as measures of machine creativity has not been established, and semantic distance tests have limited validity as predictors of human creativity as it stands (see Section 2).

Our Contributions To address these problems, we carry out the first systematic and large-scale study assessing the effectiveness of semantic distance tests for predicting creative achievement in LLMs. We measure creative achievement using six benchmarks that capture three target constructs: (i) creative writing, (ii) divergent thinking, and (iii) scientific ideation. Following careful psychometric practice, we evaluate each test on two criteria: *validity* (raw correlation r with each benchmark) and *specificity* (semi-partial $r|g$ after residualizing benchmark scores on a capability proxy g), and further derive an upper bound on the maximum attainable specificity a test can achieve as a function of its validity and its benchmark's correlation with general capabilities, which we refer to as the *validity-specificity frontier.* In summary, we make the following contributions:

1. We conduct a large-scale, systematic study evaluating the effectiveness of semantic distance tests for predicting the creative achievement of LLMs.

2. We introduce evaluation metrics which measure a test's predictive power, both in raw correlations with benchmarks (construct *validity*) and independent of what general capabilities already predict (*specificity*).

3. Our empirical findings indicate that the Divergent Association Task (DAT) is the best predictor of creative writing, the Conditional Divergent Association Task (CDAT) is the best predictor of divergent thinking, and none of the tests is a reliable predictor of scientific ideation ability.

4. We prove an upper bound on the maximum attainable specificity a creativity test can achieve, as a function of its validity and its benchmark's correlation with general capabilities (the *validity-specificity frontier*), and find that semantic distance tests are far below the frontier on nearly all benchmarks.

Overall, our findings provide practical guidance on which constructs semantic distance tests are—and are not—able to measure, while indicating there is meaningful room for future work to design novel tests with superior predictive power.

DAT. Generate 10 nouns maximally different from each other.

Response: ocean, mathematics, hammer, justice, molecule, symphony, volcano, laughter, friction, taxonomy

CDAT. Generate 10 nouns associated with **"rock"** that are maximally different from each other.

Response: stone, guitar, music, geology, cliff, mineral, foundation, cradle, concert, pebble

PACE. Starting with the seed **"rock"**, produce three 20-word association chains in which each word associates only with the previous word.

Response (chain 1 of 3): rock → stone → pebble → beach → sand → hourglass → time → clock → alarm → fire → smoke → cigarette → tobacco → leaf → tree → bark → dog → collar → shirt

Figure 2. **Example prompts and responses for each semantic distance test.** The DAT prompts for maximally distant nouns, the CDAT prompts for maximally distant nouns that are each relevant to a cue, and PACE prompts models to freely associate sequences of nouns in a 20-word chain.

2. Background and Motivating Problems

In recent years, semantic distance tests have been widely adopted to assess whether LLMs are capable of making distant associations (Chen & Ding, 2023; Cropley, 2023; Bellemare-Pepin et al., 2024). Furthermore, various strategies, such as varying sampling parameters, instructing LLMs to assume the personas of distinguished creative individuals (Wang et al., 2025), prompt engineering, and offering explicit test-taking strategies (Bellemare-Pepin et al., 2024), have all been explored to improve semantic distance test scores. Additionally, semantic distance tests are already being used to make direct claims that LLMs are *more* or *less* creative than humans (Chen & Ding, 2023; Bellemare-Pepin et al., 2024; Wang et al., 2025), which presupposes that the tests measure the same construct in both populations. However, this is problematic for several reasons.

For one, **the validity of semantic distance tests as measures of *machine* creativity has not been established.** Measures of human intelligence are often inappropriate to administer naively to LLMs (Chollet, 2019) because such tests do not inherently possess construct validity for machines.[1] A psychometric test is a valid measure of creativity

[1]The digit span test is commonly used to assess human intelligence (Wechsler, 1940), but would be inappropriate to administer to machines, where working memory is not a bottleneck to intelligence the way it is among humans. Analogously, semantic distance tests can be "hackable" by computational mechanisms that would not be considered "creative" (Boden, 2004). Consider that Transformers have been shown to implement algorithmic circuits in their weights (Olsson et al., 2022; Nanda et al., 2023), and that the DAT itself can be trivially solved by a simple algorithm

if scores correlate strongly with quantitative measures of creative achievement *within that same population*. Semantic distance tests were designed for human populations, and their external validity as measures of *machine* creativity has not yet been established.

Furthermore, **the validity of semantic distance tests as measures of *human* creativity is itself only modestly established.** Even within human populations, the external validity of these tests rests on a short and modest chain of correlations. Both the DAT (Olson et al., 2021) and forward flow[2] (Gray et al., 2019) are validated by correlation with the Alternative Uses Test (AUT): the DAT at $r \approx .32$–$.51$ (Olson et al., 2021)[3] and forward flow at $r \approx .43$–$.49$ (Beaty et al., 2021). However, meta-analyses have revealed that AUT scores themselves correlate with self-reported creative achievement at only $r \approx .17$ (Said-Metwaly et al., 2024) to $r \approx .22$ (Kim, 2008), and Baer (2011) reports near-zero correlations between AUT scores and creative achievement, casting doubt on the external validity of semantic distance tests in predicting creative outcomes among humans.

Finally, **creativity tests should measure something independent of what general capability already predicts.** A semantic distance test that correlates with, e.g., creative writing rankings, does not, on its own, establish that the test measures "creativity." A creativity test should predict creative achievement statistically *independent of* what general capability already predicts—otherwise it reduces to a measure of general capability rather than creativity. Among humans, where divergent thinking test scores are known to be confounded by general intelligence up to $r = 0.37$ (Gerwig et al., 2021), the standard practice is to factor out confounding variables (e.g., through hierarchical regression or factor analysis) before claiming a test measures "creativity" (Beaty et al., 2021). Later in this work (Section 4), we propose a similar methodology for measuring the predictive power of creativity tests independent of what general model capabilities already predict.

3. Preliminaries

With these problems in mind, we start by introducing the semantic distance tests and large-scale benchmarks we use for evaluation in this work.

over embedding distances that exceeds mean human and LLM scores (Appendix D).

[2]PACE (Qiu & Hu, 2025) uses the same scoring mechanism as forward flow.

[3]The DAT is also validated against the Bridge-the-Associative-Gap Test (Gianotti et al., 2001), although this is a measure of convergent thinking.

3.1. Semantic Distance Tests

Divergent Association Task (DAT) Introduced by Olson et al. (2021), the DAT asks subjects to name 10 nouns[4] as different from each other as possible, and scores the average cosine distance between all word pairs:

$$\text{DAT}_{\mathcal{E}}(W) := \tfrac{100}{n(n-1)} \sum_{i<j}^{N} (1 - \cos_{\mathcal{E}}(\mathbf{w}_i, \mathbf{w}_j)) \quad (1)$$

In addition to computing scores under the standard GloVe 840B embeddings (Pennington et al., 2014) used in Olson et al. (2021), we test robustness under multiple embedding models in Section 4.

Conditional Divergent Association Task (CDAT) Creative artifacts should be both novel *and* useful (Varshney, 2019; Boden, 2004; Simonton, 2004; Maher, 2010). Nakajima et al. (2026) argued that the original DAT failed to measure utility, and proposed the conditional DAT (CDAT), in which a cue word c is added, and each word must be sufficiently relevant to the cue word in order for its novelty to count. For words $W = \{w_1, \ldots, w_n\}$ generated for cue c, novelty and appropriateness are measured according to:

$$\text{CDAT-N}_{\mathcal{E}}(W) := \tfrac{100}{n(n-1)} \sum_{i<j}^{N} (1 - \cos_{\mathcal{E}}(\mathbf{w}_i, \mathbf{w}_j)) \quad (2)$$

$$\text{CDAT-A}_{\mathcal{E}}(W \mid c) := \tfrac{100}{n} \sum_{i=1}^{n} \cos_{\mathcal{E}}(\mathbf{c}, \mathbf{w}_i) \quad (3)$$

Note that Equation (2) is equivalent to Equation (1), but that CDAT-N and DAT scores differ due to cue-based prompting under the CDAT.

Parallel Association Chain Evaluation (PACE) PACE (Qiu & Hu, 2025), inspired by forward-flow associative-chain methods (Gray et al., 2019), prompts models to produce three parallel 20-word association chains from a seed, and the score is the mean cumulative cosine distance along each chain. For a chain $C = (w_1, \ldots, w_L)$:

$$\text{PACE}_{\mathcal{E}}(C) := \tfrac{1}{L-1} \sum_{i=2}^{L} \tfrac{1}{i-1} \sum_{j=1}^{i-1} (1 - \cos_{\mathcal{E}}(\mathbf{w}_i, \mathbf{w}_j)) \quad (4)$$

The model-level score is the mean of $\text{PACE}_{\mathcal{E}}(C)$ across three chains per seed and across all seeds. In the original paper, PACE correlated with Chatbot Arena CW at $\rho \approx 0.74$ with $n = 30$. The metric was originally tested under FastText embeddings (Mikolov et al., 2018), although Qiu & Hu (2025) also saw significant rank correlations with Chatbot Arena CW under multiple distinct embedding models.

[4]Usually, only the first seven valid nouns are scored.

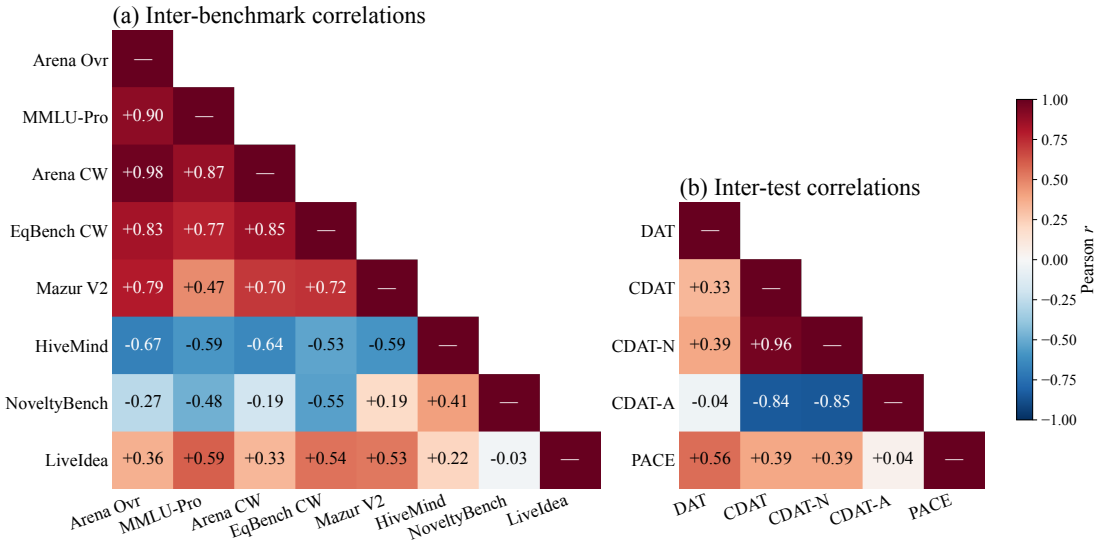

Figure 3. **Inter-benchmark and inter-test correlations. (a)** Benchmarks are ordered by construct: general capabilities (Arena Ovr, MMLU-Pro), creative writing (Arena CW, EqBench CW, Mazur CW), divergent thinking (Hivemind, NoveltyBench), and scientific ideation (LiveIdea). n per cell ranges from 8 (Mazur $\times$ Hivemind) to 54. **(b)** Inter-test correlations among DAT, CDAT, CDAT-N, CDAT-A, PACE using composite scores across GloVe, FastText, and SBERT.

3.2. Creative Achievement Benchmarks

We evaluate each semantic distance test against three target constructs (creative writing, divergent thinking, and scientific ideation) spanning six large-scale benchmarks.

3.2.1. Creative Writing

Chatbot Arena Creative Writing (Arena CW) (Chiang et al., 2024). Arena CW is a creative writing benchmark where models respond to open-ended, user-submitted creative writing prompts, and user preference ratings are aggregated across pairwise evaluations between models, leading to per-model Elo ratings. Arena CW is the most capability-loaded of the three creative writing benchmarks ($r = 0.98$ with Arena Overall; Figure 3).

EQ-Bench Creative Writing (Paech, 2023). EQ-Bench is a creative writing benchmark where models produce responses to a set of 32 distinct prompts designed to challenge models in areas like humor, romance, spatial awareness, and unique perspectives. Responses are scored head-to-head by Claude Sonnet using a 9-criterion rubric and aggregated into Elo ratings, and the benchmark incorporates strategies to reduce known LLM-as-a-judge biases (length, position, poetic incoherence, etc.). EQ-Bench correlates strongly with Arena Overall ($r = 0.83$; Figure 3), although less than Arena CW.

Mazur Creative Writing (Mazur, 2025). Mazur Creative Writing is a creative writing benchmark that tests how well LLMs incorporate a set of mandatory story elements (characters, objects, core concepts, attributes, motivations, etc.) in a short creative story. An ensemble of LLM judges applies an ∼18-question rubric assessing whether mandatory elements are adequately represented, as well as the overall narrative quality. The final story score is the mean of the per-grader scores. Mazur Creative Writing correlates strongly with Arena Overall at $r = 0.79$, although less than both Arena CW and EQ-Bench.

All three are heavily capability-loaded, having correlations $r = 0.98/0.83/0.79$ with Arena Overall, respectively, motivating the use of both validity *and* specificity as evaluation criteria (see Section 4).

3.2.2. Divergent Thinking

We use two divergent thinking benchmarks, each of which measures output diversity in LLM responses to open-ended prompts.

Hivemind (Jiang et al., 2025). Hivemind is a divergent thinking benchmark which measures output diversity across open-ended queries, drawn from real-world user-ChatGPT interactions. Categories include brainstorm and ideation ("Suggest a feature for a smartwatch designed specifically for senior citizens."), philosophical questions ("How do I understand what I want?"), speculative and hypothetical scenarios ("Create a short review of a future movie."), and ambiguous everyday questions ("How can I live on $1,000 per month?"). Scoring measures intra-model repetition via average pairwise embeddings dissimilarity[5] using OpenAI's

[5]The original paper uses similarity, but for consistency with the divergent thinking construct, we report its complement.

`text-embedding-3-small` model. Hivemind scores are negatively correlated with capability ($r = -0.67$ with Arena Overall, $r - 0.59$ with MMLU-Pro).

NoveltyBench (Zhang et al., 2025). NoveltyBench evaluates the output diversity of LLMs on a similar collection of real-world user-ChatGPT interactions, as well as open-ended prompts across four categories: randomness ("Roll a make-believe 20-sided die"), factual information ("List a capital city in Africa."), creative text writing ("Tell me a riddle."), and subjectivity ("What's the best car to get in 2023?"). Scoring assesses the extent to which a sequence of generations is both diverse and high-quality, using a cumulative utility measure that penalizes functional equivalence to prior responses. NoveltyBench is the most capability-independent benchmark, obtaining $r = -0.27$ with Arena Overall and $r = -0.48$ with MMLU-Pro.

3.2.3. Scientific Ideation

LiveIdeaBench (Ruan et al., 2026). LiveIdeaBench assesses LLMs on open-ended scientific idea generation. Given a single keyword (e.g., a domain term) drawn from a set spanning 18 scientific disciplines, a model must propose a research idea relevant to that keyword in a brief response. Each generated idea is scored by a panel of LLM judges along five dimensions—*fluency* (does the response actually contain a usable idea), *feasibility*, *clarity*, *originality*, and *flexibility*—with overall scores reported as the average across all dimensions. The original benchmark explicitly demonstrates that scientific ideation can dissociate from general capability (e.g., QwQ-32B-preview outperforms much larger frontier models on Originality despite being weaker on knowledge tests), and empirically, we observe a similar trend where LiveIdeaBench is only moderately correlated with general capabilities (Figure 3a; $r = 0.36$ on Arena Overall, $r = 0.59$ on MMLU-Pro).

4. Evaluation Method

4.1. Models and Inference

In order to evaluate the effectiveness of semantic distance tests for predicting creative achievement benchmarks, we obtain raw test scores across a large set ($n = 54$) of instruction-tuned LLMs across 10 providers (OpenAI, Anthropic, Google, Meta, Mistral, Qwen, DeepSeek, Cohere, NVIDIA, Microsoft) via OpenRouter. DAT and CDAT use $T \in \{1.0, 1.5, 2.0\}$ (40 trials per temperature, with `top_p`= 1, `top_k`= 0). PACE follows the original setup in Qiu & Hu (2025), where sampling is done with temperature $T = 0$ and stochasticity is controlled by varying the random seed for each anchor word. We use 50 anchor words, and collect three parallel 20-word chains per seed. Lastly, on the CDAT, we use 50 cue words, and following Nakajima et al. (2026), per-cue appropriateness values are compared to a random-noun baseline via Welch's t-test, with a Benjamini–Hochberg false discovery rate (FDR) correction at $\alpha = .001$ across models within each temperature. A model's responses at a given temperature are retained if their FDR-adjusted p-value passes and its mean appropriateness exceeds the random baseline. The CDAT score we report is the mean of CDAT-N across passing temperatures. For reproducibility purposes, exact prompts used for each test are given in Appendix E.

Embedding models Each semantic distance test score depends on the embedding model used for pairwise cosine similarity. To evaluate robustness to embedding choice, we score each test under three embeddings—GloVe 840B 300d, FastText crawl-300d-2M, and Sentence-BERT `all-mpnet-base-v2`—which differ in training. GloVe and FastText are static word-level co-occurrence models (300-dim), while Sentence-BERT (`all-mpnet-base-v2`; 768-dim) is a transformer sentence encoder trained via contrastive sentence-pair objectives.

4.2. Evaluation Metrics

We evaluate each test on two criteria, validity and specificity, each of which is motivated and defined below.

Validity Standard psychometric criteria used in human creativity research (Beaty et al., 2021; Olson et al., 2021) indicate that a test has construct validity if its scores correlate with external measures of that construct. Therefore, we report *validity* as the raw Pearson correlation between a test score X and benchmark score Y, given by $r(X, Y)$.

Specificity Since creative achievement benchmarks are themselves highly capability-loaded (for example, Arena CW correlates with Arena Overall at $r = 0.98$) a high raw $r(X, Y)$ may simply reflect that the test tracks general capability. We therefore report *specificity*, the semi-partial Pearson correlation between the test score X and the benchmark Y residualized on a capability stack g, given by $r(X, Y \mid g) := r\Big(X,\ Y - \hat{Y}_g\Big)$. Here, $\hat{Y}_g$ is the ordinary least squares prediction of Y from Arena Overall Elo (preference-based) and MMLU-Pro accuracy (Wang et al., 2024) (knowledge- and reasoning-based). The semi-partial correlation measures to what extent the test predicts the benchmark's capability-controlled variance, with a significant positive $r(X, Y \mid g)$ indicating the test predicts variance that capability cannot already explain.

For both validity and specificity, we report two-sided p-values from the standard Pearson t-test, $t = r\sqrt{(n-2-k)/(1-r^2)} \sim t_{n-2-k}$, where k is the number of controls regressed out ($k = 0$ for validity and $k = 2$

	Creative Writing						*Divergent Thinking*				*Scientific Ideation*	
	Arena CW ↑ n=54/40		**EQ-Bench CW** ↑ n=35/27		**Mazur CW** ↑ n=21/18		**Hivemind Div.** ↑ n=25/21		**NovBench Util.** ↑ n=14/10		**LiveIdeaBench** ↑ n=17/17	
	Valid.	Spec.	Valid.	Spec.	Valid.	Spec.	Valid.	Spec.	Valid.	Spec.	Valid.	Spec.
Overall (mean z-score across 3 embeddings)												
DAT	$\mathbf{+.47}^{***}$	$+.05$	$\mathbf{+.72}^{***}$	$\mathbf{+.41}^{*}$	$\mathbf{+.60}^{**}$	$+.49$	$+.33$	$+.01$	$+.15$	$-.21$	$-.01$	$+.24$
CDAT	$-.13$	$+.22$	$-.06$	$+.03$	$+.07$	$+.43$	$+.25$	$+.10$	$+.60$	$+.60$	$+.03$	$+.21$
CDAT-N	$-.18$	$+.20$	$-.14$	$+.08$	$+.09$	$+.40$	$+.24$	$+.08$	$\mathbf{+.54}^{*}$	$+.45$	$-.11$	$+.02$
CDAT-A	$\mathbf{+.54}^{***}$	$-.05$	$\mathbf{+.48}^{**}$	$+.05$	$+.24$	$-.28$	$-.39$	$-.09$	$\mathbf{-.67}^{**}$	$-.40$	$+.20$	$+.08$
PACE	$\mathbf{+.71}^{***}$	$+.11$	$\mathbf{+.71}^{***}$	$+.21$	$\mathbf{+.76}^{***}$	$+.14$	$-.05$	$+.33$	$+.18$	$-.00$	$+.11$	$-.00$
GloVe 840B												
DAT	$\mathbf{+.37}^{**}$	$-.01$	$\mathbf{+.57}^{***}$	$+.28$	$+.43$	$+.39$	$+.21$	$+.04$	$-.22$	$-.46$	$-.20$	$+.04$
CDAT	$-.07$	$+.14$	$-.03$	$+.05$	$+.43$	$+.44$	$+.35$	$+.22$	$+.56$	$+.53$	$-.06$	$+.18$
CDAT-N	$-.20$	$+.11$	$-.16$	$+.04$	$+.16$	$+.46$	$+.26$	$+.10$	$+.49$	$+.30$	$-.15$	$-.02$
CDAT-A	$\mathbf{+.54}^{***}$	$-.00$	$\mathbf{+.49}^{**}$	$+.04$	$+.17$	$-.34$	$\mathbf{-.40}^{*}$	$-.09$	$\mathbf{-.61}^{*}$	$-.30$	$+.20$	$+.10$
PACE	$\mathbf{+.72}^{***}$	$+.00$	$\mathbf{+.70}^{***}$	$+.15$	$\mathbf{+.71}^{***}$	$+.12$	$-.13$	$+.22$	$+.10$	$-.17$	$-.02$	$-.16$
FastText crawl-300d-2M												
DAT	$\mathbf{+.35}^{**}$	$+.00$	$\mathbf{+.59}^{***}$	$+.29$	$\mathbf{+.64}^{**}$	$\mathbf{+.57}^{*}$	$+.32$	$-.11$	$+.32$	$+.26$	$-.06$	$+.16$
CDAT	$-.13$	$+.35$	$-.08$	$+.04$	$-.04$	$+.38$	$+.23$	$+.05$	$+.63$	$+.62$	$+.10$	$+.29$
CDAT-N	$-.13$	$+.32$	$-.10$	$+.05$	$+.01$	$+.34$	$+.21$	$+.04$	$\mathbf{+.54}^{*}$	$+.46$	$-.04$	$+.12$
CDAT-A	$\mathbf{+.51}^{***}$	$-.09$	$\mathbf{+.45}^{**}$	$+.05$	$+.30$	$-.23$	$\mathbf{-.41}^{*}$	$-.08$	$\mathbf{-.70}^{**}$	$-.45$	$+.17$	$+.03$
PACE	$\mathbf{+.72}^{***}$	$+.29$	$\mathbf{+.71}^{***}$	$+.25$	$\mathbf{+.71}^{***}$	$+.08$	$-.13$	$+.26$	$+.19$	$+.07$	$+.13$	$+.08$
Sentence-BERT all-mpnet-base-v2												
DAT	$\mathbf{+.44}^{***}$	$+.13$	$\mathbf{+.63}^{***}$	$\mathbf{+.42}^{*}$	$\mathbf{+.44}^{*}$	$+.29$	$+.25$	$+.11$	$+.11$	$-.12$	$+.29$	$+.37$
CDAT	$-.08$	$+.24$	$-.08$	$-.00$	$+.33$	$+.34$	$+.27$	$+.14$	$+.60$	$+.62$	$-.05$	$+.16$
CDAT-N	$-.21$	$+.13$	$-.15$	$+.14$	$+.08$	$+.37$	$+.23$	$+.10$	$\mathbf{+.57}^{*}$	$+.50$	$-.16$	$-.04$
CDAT-A	$\mathbf{+.54}^{***}$	$-.04$	$\mathbf{+.48}^{**}$	$+.05$	$+.24$	$-.25$	$-.37$	$-.10$	$\mathbf{-.68}^{**}$	$-.42$	$+.22$	$+.11$
PACE	$\mathbf{+.60}^{***}$	$+.05$	$\mathbf{+.65}^{***}$	$+.23$	$\mathbf{+.76}^{***}$	$+.18$	$+.18$	$\mathbf{+.47}^{*}$	$+.24$	$+.09$	$+.17$	$+.04$

Table 1. **Validity and specificity for every test–benchmark pair.** The **Overall** block (top) averages across the three embeddings and is the main result referenced in the text. For robustness, we also report per-embedding blocks. Green cells mark the best test per benchmark in the Overall block, bold = significant at $p < .05$, while stars denote significance levels ($^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$). Lastly, sample sizes n = val/spec are reported under each header, and specificity has reduced sample sizes due to marginally lower model overlap.

for Arena Overall + MMLU-Pro used for specificity).

4.3. The Validity-Specificity Frontier

What is the maximum attainable specificity a test can achieve, as a function of its validity and its benchmark's correlation with general capabilities? We refer to this as the *validity-specificity frontier*, and prove a bound on this quantity below.

Theorem 1 (Validity-Specificity Frontier)**.** *Let $g = (Z_1, \ldots, Z_k)$ be a vector of general capability proxies, let $\hat{Y}_g$ denote the ordinary least squares prediction of Y from g, and define $R = \mathrm{corr}(Y, \hat{Y}_g)$. For any test X with validity $v = \mathrm{corr}(X, Y)$, the semi-partial correlation $r(X, Y \mid g) := r(X,\, Y - \hat{Y}_g)$ satisfies*

$$|r(X, Y \mid g)| \;\le\; v\sqrt{1 - R^2} \;+\; |R|\,\sqrt{1 - v^2}. \qquad (5)$$

The proof of Theorem 1 is given in Section A of the appendix. The intuition is that a creativity test highly correlated with a capability-loaded benchmark *cannot* be very decoupled from capability (i.e., specificity). In particular, a perfectly valid test ($v = 1$) has specificity bounded by $\sqrt{1 - R^2}$, and a benchmark with $|R|$ near 1 therefore leaves almost no specificity headroom for *any* test, regardless of construction.

This is why Arena CW ($R = 0.98$ on $g = (\text{Arena Overall}, \text{MMLU-Pro})$) caps specificity at ≈ 0.20 for a perfect-validity test, while NoveltyBench Utility ($R \approx -0.33$) admits a much wider ceiling (Figure 4, bottom row).

5. Results

In Table 1, we report the validity and specificity across all tests and benchmarks. Test and benchmark scores used for correlations are fully reported in Table 2 and Table 3, respectively. Figure 4 visualizes validity and specificity aggregated across the three constructs and plots the validity-specificity frontier we derived in Section 4.3.

Evaluating specificity reveals that PACE is mostly a capability proxy. A core motivation addressed in Section 2

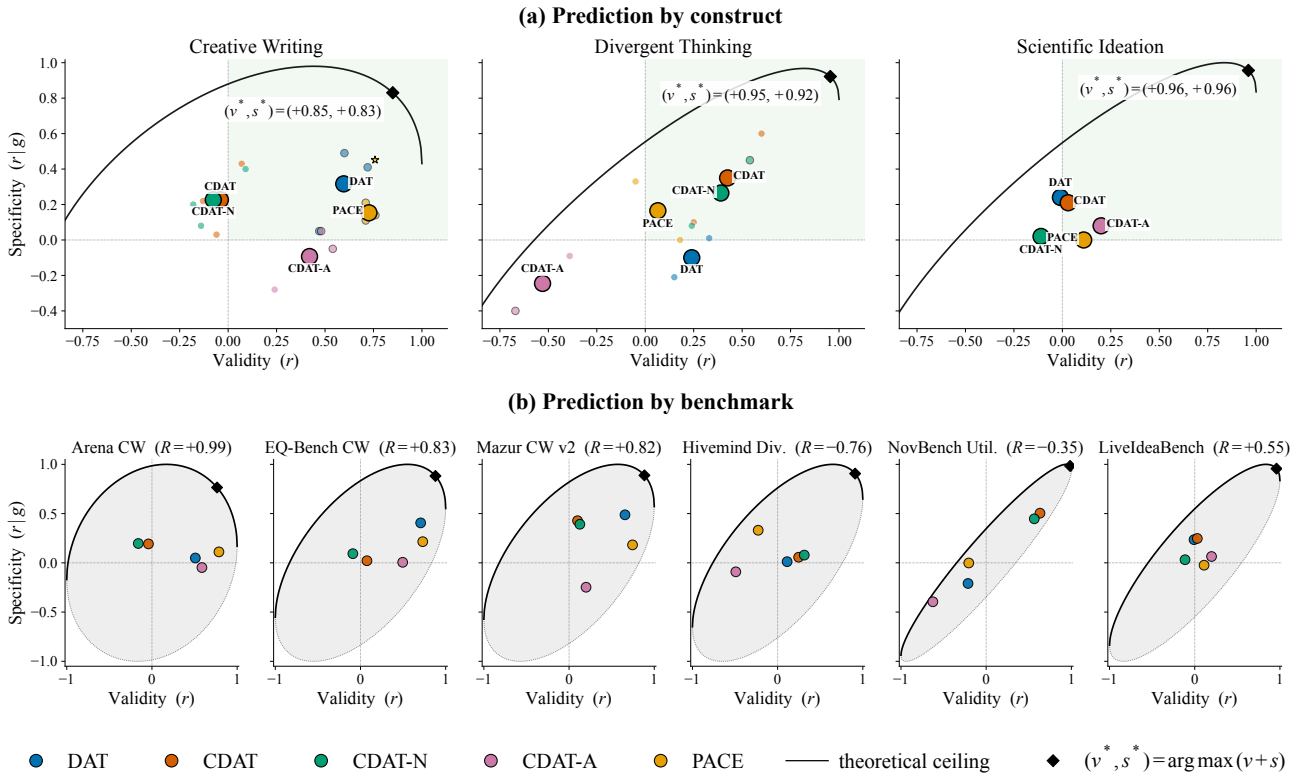

Figure 4. **Validity and specificity by construct and benchmark, with theoretical ceilings.** *(a) Prediction by construct:* Each small translucent point is a (test, benchmark) cell from the Overall block of Table 1, while the large black-outlined circle is the construct-level average across all benchmarks. A gold ★ = significant on both axes. The black curve is the construct-level theoretical specificity ceiling — the unweighted mean across the panel's benchmarks of the per-benchmark bound $|r(X, Y \mid g)| \leq v\sqrt{1-R^2} + |R|\sqrt{1-v^2}$ obtained in Theorem 1. Test effectiveness varies significantly by construct, with no single test predicting all constructs well. *(b) Prediction by benchmark:* per-benchmark specificity-ceiling lenses with the panel's R in the title. The grey region is the feasible $(v, r(X, Y \mid g))$ set and the solid black curve is the upper ceiling. The colored dots are the six tests. Observed tests sit well inside the lens in every panel, indicating considerable headroom to design better creativity tests.

is that tests should measure aspects of creative achievement independent of what general capabilities already predict. We measure this via the specificity metric defined in Section 4. On creative writing benchmarks, PACE obtains strong and statistically significant validity ($r \in [0.71, 0.76]$); however, after controlling for general capabilities, PACE collapses to non-significant specificity ($r|g \in [+0.11, +0.21]$). On creative writing, the chained-association testing methodology largely measures model qualities that capability already predicts.

The Divergent Association Task (DAT) is the best predictor of creative writing. DAT's validity is moderate across the three creative writing benchmarks (Arena CW $r = +0.47$, EQ-Bench CW $r = +0.72$, Mazur CW $r = +0.60$), and the EQ-Bench and Mazur cells retain meaningful specificity once capability is partialled out ($r|g = +0.41^{*}$ and $+0.49$). As mentioned, PACE has high raw validity averaged across the three creative writing benchmarks ($\approx +0.73$), but its specificity averages only $+0.15$. Lastly, CDAT's appropriateness facet (CDAT-A) has high validity ($r = +0.54^{***}, +0.48^{**}, +0.24$) but non-positive specificity throughout, predicting rankings only because it tracks capability.

The Conditional DAT is the best predictor of divergent thinking. The CDAT is the only test whose specificity is positive on both divergent thinking benchmarks (Hivemind $r|g = +0.10$; NoveltyBench Utility $r|g = +0.60$, $n = 8$). Furthermore, its appropriateness facet, CDAT-A, has both negative validity and specificity for divergent thinking (NoveltyBench $r = -0.67^{**}$, $r|g = -0.40$), consistent with appropriateness being a convergent-thinking measure that should anti-correlate with output diversity by construction.

None of the tests is a good predictor of scientific ideation. On LiveIdeaBench ($n = 17$), every test's raw validity is near zero ($r \in [-0.11, +0.20]$) and no specificity cell reaches $p < 0.05$. DAT and CDAT show modest positive specificity ($r|g \approx +0.24$ and $+0.21$), but neither is statistically significant. We discuss why semantic distance alone may be insufficient to predict scientific ideation ability in Section 6.

No single test predicts all constructs well. With DAT being the best predictor of creative writing, CDAT the best predictor of divergent thinking, and none of the tests a good predictor of scientific ideation, it is evident that each test is a local predictor of creative achievement tied to a specific construct rather than a general-purpose "creativity" measure.

How close are tests to the validity-specificity frontier? Empirically, we find that observed tests are well below the validity-specificity frontier for nearly every benchmark in Figure 4(b). This indicates significant headroom to design more effective creativity tests. The $(v^\star, s^\star)$ points marked in Figure 4 give the validity-specificity pair on the frontier that maximizes the sum $v + s$ for each construct or benchmark. They are the most that any test could jointly achieve on both axes given the benchmark's coupling to capability, and the gap between each observed test and its $(v^\star, s^\star)$ measures how much room is left for better tests of the same construct.

6. Discussion

Designing creativity tests that predict scientific ideation ability In Section 5, we found that existing semantic distance tests fail to reliably predict scientific ideation ability. Most surprisingly, even the CDAT, which measures both utility and novelty, fails to achieve significant validity and specificity on this construct. Future evaluations of this ability may need to better reflect the cognitive primitives implicated in scientific ideation, and target specific mechanical bottlenecks that limit LLM creative achievement, rather than rely solely on semantic distance measures.

A recent line of work has designed creativity tests more carefully around the cognitive processes that underlie scientific discovery. For example, Nagarajan et al. (2025) proposed algorithmic tasks that required creative, far-sighted leaps, mirroring cognitive mechanisms implicated in the scientific process (Boden, 2004; Koestler, 1964). Later, Schapiro et al. (2025b) proposed a constrained graph pathfinding task, where constraints reflected common pitfalls observed in large-scale LLM-for-science studies (Si et al., 2024; 2025). However, both designs required training models from scratch, and were limited by the synthetic nature of the tasks. Addressing both of these problems, Wadhwa et al. (2026) proposed a test-time creativity evaluation (CREATE) that measured associative creativity using real knowledge graphs, scoring an LLM by the diversity and validity of multi-hop associations it generated between concepts drawn from those graphs. As we have shown in this study, future tests that claim to evaluate aspects of scientific "creativity" should be validated by specificity and validity measures on real-world benchmarks.

Testing for transformational creativity One facet of creative achievement understudied by the existing testing and benchmarking paradigm is transformational creativity (Boden, 2004), which has historically played an outsized role in economic innovation, scientific discovery, and artistic achievement. Transformational creativity involves altering existing conceptual spaces in ways that radically transform the scope of conceivable artifacts.[6] Recent work has begun to explore both empirical evaluation of this ability (Sun et al., 2025), as well as its conceptual and theoretical foundations (Schapiro et al., 2025a), each of which should be incorporated in future large-scale benchmarking and test design studies.

Limitations of tests versus open-world evaluations Although creativity tests offer convenient and controllable scoring, there are inherent limitations to how much variance in creative achievement minimal assessments can capture, due to the complex, long-horizon, and open-ended nature of real-world tasks. Kapoor et al. (2026) has recently called for *open-world evaluations*: "long-horizon, messy, real-world tasks assessed through small-sample qualitative analysis rather than benchmark-scale automation." Although in some respects, creativity can be reduced to cognitive primitives and concrete mechanisms (Simonton, 2023), real-world creative processes are complex and typically modeled using distinct stages (Wallas, 1926). Accordingly, creative achievement may be best measured through a combination of minimal tests and open-ended, long-horizon assessments.

6.1. Limitations

Before concluding, several limitations of our work are worth discussing. Some benchmarks in our suite have $n < 25$ models with full coverage of the capability stack due to limited coverage on already-evaluated models (see Table 1). Validity and specificity should be interpreted cautiously here, and future work can aim to expand benchmark coverage to ensure greater statistical reliability. Although we study a large set of LLMs, this pool consists exclusively of instruction-tuned LLMs, and our findings may not generalize to base models.

7. Conclusion

In this work, we conducted the first systematic study to assess the effectiveness of semantic distance tests in predicting creative achievement across three constructs: creative writing, divergent thinking, and scientific ideation. Our findings provide clear practical takeaways and directions for future work, identifying tests which are best suited to predict each construct, and suggesting that novel tests are needed to reliably predict scientific ideation ability.

[6] Einstein's relativity theory and Copernicus's heliocentric theory being two canonical examples (Schapiro et al., 2025a).

References

Baer, J. How divergent thinking tests mislead us: Are the torrance tests still relevant in the 21st century? the division 10 debate. *Psychology of Aesthetics, Creativity, and the Arts*, 5(4):309, 2011.

Beaty, R. E., Zeitlen, D. C., Baker, B. S., and Kenett, Y. N. Forward flow and creative thought: Assessing associative cognition and its role in divergent thinking. *Thinking Skills and Creativity*, 41, 9 2021. ISSN 18711871. doi: 10.1016/j.tsc.2021.100859.

Bellemare-Pepin, A., Lespinasse, F., Thölke, P., Harel, Y., Mathewson, K., Olson, J. A., Bengio, Y., and Jerbi, K. Divergent Creativity in Humans and LLMs. Technical report, 2024.

Boden, M. A. *The Creative Mind: Myths and Mechanisms*. Routledge, 2004.

Chen, H. and Ding, N. Probing the Creativity of Large Language Models: Can models produce divergent semantic association? 10 2023. URL http://arxiv.org/abs/2310.11158.

Chiang, W.-L., Zheng, L., Sheng, Y., Angelopoulos, A. N., Li, T., Li, D., Zhang, H., Zhu, B., Jordan, M., Gonzalez, J. E., et al. Chatbot arena: An open platform for evaluating llms by human preference. *arXiv preprint arXiv:2403.04132*, 2024.

Chollet, F. On the measure of intelligence. *arXiv preprint arXiv:1911.01547*, 2019.

Cropley, D. Is artificial intelligence more creative than humans?: Chatgpt and the divergent association task. *Learning Letters*, 2:13–13, 2023.

Gerwig, A., Miroshnik, K., Forthmann, B., Benedek, M., Karwowski, M., and Holling, H. The relationship between intelligence and divergent thinking—a meta-analytic update. *Journal of Intelligence*, 9(2):23, 2021.

Gianotti, L. R., Mohr, C., Pizzagalli, D., Lehmann, D., and Brugger, P. Associative processing and paranormal belief. *Psychiatry and clinical neurosciences*, 55(6):595–603, 2001.

Gray, K., Anderson, S., Chen, E. E., Kelly, J. M., Christian, M. S., Patrick, J., Huang, L., Kenett, Y. N., and Lewis, K. "Forward flow": A new measure to quantify free thought and predict creativity. *American Psychologist*, 74(5):539, 2019.

Jiang, L., Chai, Y., Li, M., Liu, M., Fok, R., Dziri, N., Tsvetkov, Y., Sap, M., Albalak, A., and Choi, Y. Artificial hivemind: The open-ended homogeneity of language models (and beyond), 2025. URL https://arxiv.org/abs/2510.22954.

Kapoor, S., Kirgis, P., Schwartz, A., Rabanser, S., Allaire, J., Bommasani, R., Dubois, M., Hadfield, G., Hall, A., Hooker, S., Lazar, S., Newman, S., Papailiopoulos, D., Tekofsky, S., Toner, H., Ududec, C., and Narayanan, A. Open-world evaluations for measuring frontier AI capabilities. https://cruxevals.com/open-world-evaluations.pdf, 2026.

Kim, K. H. Meta-analyses of the relationship of creative achievement to both iq and divergent thinking test scores. *The Journal of Creative Behavior*, 42(2):106–130, 2008.

Koestler, A. *The Act of Creation*. Macmillan, 1964.

Maher, M. L. Evaluating creativity in humans, computers, and collectively intelligent systems. In *Proceedings of the 1st DESIRE Network Conference on Creativity and Innovation in Design*, DESIRE '10, pp. 22–28, 2010.

Mazur, L. LLM creative story-writing benchmark (v2). https://github.com/lechmazur/writing, 2025.

Mednick, S. The associative basis of the creative process. *Psychological Review*, 69(3):220, 1962.

Mikolov, T., Grave, E., Bojanowski, P., Puhrsch, C., and Joulin, A. Advances in pre-training distributed word representations. In *Proceedings of the eleventh international conference on language resources and evaluation (LREC 2018)*, 2018.

Nagarajan, V., Wu, C. H., Ding, C., and Raghunathan, A. Roll the dice & look before you leap: Going beyond the creative limits of next-token prediction. arXiv:2504.15266, 2025.

Nakajima, K., Zuiderveld, J., and Pezzelle, S. Beyond divergent creativity: A human-based evaluation of creativity in large language models. *arXiv preprint arXiv:2601.20546*, 2026.

Nanda, N., Chan, L., Lieberum, T., Smith, J., and Steinhardt, J. Progress measures for grokking via mechanistic interpretability. *arXiv preprint arXiv:2301.05217*, 2023.

Olson, J. A., Nahas, J., Chmoulevitch, D., Cropper, S. J., and Webb, M. E. Naming unrelated words predicts creativity. 4 2021. doi: 10.1073/pnas.2022340118/-/DCSupplemental.y.

Olsson, C., Elhage, N., Nanda, N., Joseph, N., DasSarma, N., Henighan, T., Mann, B., Askell, A., Bai, Y., Chen, A., Conerly, T., Drain, D., Ganguli, D., Hatfield-Dodds, Z., Hernandez, D., Johnston, S., Jones, A., Kernion, J., Lovitt, L., Ndousse, K., Amodei, D., Brown, T., Clark, J., Kaplan, J., McCandlish, S., and Olah, C. In-context learning and induction heads, 2022. URL https://arxiv.org/abs/2209.11895.

Paech, S. J. Eq-bench: An emotional intelligence benchmark for large language models. *arXiv preprint arXiv:2312.06281*, 2023.

Pennington, J., Socher, R., and Manning, C. D. Glove: Global vectors for word representation. In *Proceedings of the 2014 conference on empirical methods in natural language processing (EMNLP)*, pp. 1532–1543, 2014.

Qiu, Z. and Hu, R. Deep associations, high creativity: A simple yet effective metric for evaluating large language models. In Christodoulopoulos, C., Chakraborty, T., Rose, C., and Peng, V. (eds.), *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pp. 10859–10872, Suzhou, China, November 2025. Association for Computational Linguistics. ISBN 979-8-89176-332-6. doi: 10.18653/v1/2025.emnlp-main.550. URL https://aclanthology.org/2025.emnlp-main.550/.

Ruan, K., Wang, X., Hong, J., Wang, P., Liu, Y., and Sun, H. Evaluating llms' divergent thinking capabilities for scientific idea generation with minimal context. *Nature Communications*, 2026.

Said-Metwaly, S., Taylor, C. L., Camarda, A., and Barbot, B. Divergent thinking and creative achievement—how strong is the link? an updated meta-analysis. *Psychology of Aesthetics, Creativity, and the Arts*, 18(5):869, 2024.

Schapiro, S., Black, J., and Varshney, L. R. Transformational Creativity in Science: A Graphical Theory. *arXiv preprint arXiv:2504.18687*, 2025a.

Schapiro, S., Shashidhar, S., Gladstone, A., Black, J., Moon, R., Hakkani-Tur, D., and Varshney, L. R. Combinatorial Creativity: A New Frontier in Generalization Abilities. 11 2025b. URL http://arxiv.org/abs/2509.21043.

Si, C., Yang, D., and Hashimoto, T. Can LLMs Generate Novel Research Ideas? A Large-Scale Human Study with 100+ NLP Researchers. 9 2024. URL http://arxiv.org/abs/2409.04109.

Si, C., Hashimoto, T., and Yang, D. The Ideation-Execution Gap: Execution Outcomes of LLM-Generated versus Human Research Ideas. 6 2025. URL http://arxiv.org/abs/2506.20803.

Simonton, D. K. *Creativity in Science: Chance, Logic, Genius, and Zeitgeist*. Cambridge University Press, 2004.

Simonton, D. K. The blind-variation and selective-retention theory of creativity: Recent developments and current status of bvsr. *Creativity Research Journal*, 35(3):304–323, 2023.

Stevenson, C., Smal, I., Baas, M., Grasman, R., and Van Der Maas, H. Putting GPT-3's Creativity to the (Alternative Uses) Test. In *International Conference on Computational Creativity*, 2022. URL http://osf.io/vmk3c/.

Sun, Y., Hu, S., Zhou, G., Zheng, K., Hajishirzi, H., Dziri, N., and Song, D. Omega: Can llms reason outside the box in math? evaluating exploratory, compositional, and transformative generalization. *arXiv preprint arXiv:2506.18880*, 2025.

Varshney, L. R. Mathematical limit theorems for computational creativity. *IBM Journal of Research and Development*, 63(1), 1 2019. ISSN 21518556. doi: 10.1147/JRD.2019.2893907.

Wadhwa, M., Roy, T. S., Lederman, H., Li, J. J., and Durrett, G. Create: Testing llms for associative creativity. *arXiv preprint arXiv:2603.09970*, 2026.

Wallas, G. *The art of thought*. Number 24. Harcourt, Brace, 1926.

Wang, D., Huang, D., Shen, H., and Uzzi, B. A large-scale comparison of divergent creativity in humans and large language models. *Nature Human Behaviour*, pp. 1–10, 2025.

Wang, Y., Ma, X., Zhang, G., Ni, Y., Chandra, A., Guo, S., Ren, W., Arulraj, A., He, X., Jiang, Z., et al. Mmlu-pro: A more robust and challenging multi-task language understanding benchmark. *Advances in Neural Information Processing Systems*, 37:95266–95290, 2024.

Wechsler, D. The measurement of adult intelligence. *The Journal of Nervous and Mental Disease*, 91(4):548, 1940.

Zhang, Y., Diddee, H., Holm, S., Liu, H., Liu, X., Samuel, V., Wang, B., and Ippolito, D. NoveltyBench: Evaluating language models for humanlike diversity. In *Conference on Language Modeling (COLM)*, 2025. URL https://arxiv.org/abs/2504.05228.

A. Derivation of the Validity-Specificity Frontier

The theoretical ceilings plotted in Figure 4 (benchmark validity-specificity frontier) are built on a covariance matrix bound that we prove here.

Theorem 1 (Validity-Specificity Frontier, restated). *Let $g = (Z_1, \ldots, Z_k)$ be a vector of general capability proxies, let $\hat{Y}_g$ denote the ordinary least squares prediction of Y from g, and define $R = \mathrm{corr}(Y, \hat{Y}_g)$. For any test X with validity $v = \mathrm{corr}(X, Y)$, the semi-partial correlation $r(X, Y \mid g) := r(X,\, Y - \hat{Y}_g)$ satisfies*

$$|r(X, Y \mid g)| \;\le\; v\sqrt{1 - R^2} \;+\; |R|\,\sqrt{1 - v^2}.$$

Proof. Without loss of generality, we standardize X and Y to have unit variance, and let $\tilde{Z} := \hat{Y}_g / R$ so that $\sigma(\tilde{Z}) = 1$ as well. Since correlation is scale-invariant, $\mathrm{corr}(X, \tilde{Z}) = \mathrm{corr}(X, \hat{Y}_g) =: a$ and $\mathrm{corr}(Y, \tilde{Z}) = R$. The covariance matrix of the unit-variance triple $(X, Y, \tilde{Z})$,

$$\Sigma \;=\; \begin{pmatrix} 1 & v & a \\ v & 1 & R \\ a & R & 1 \end{pmatrix},$$

must be positive semi-definite. Expanding the determinant gives

$$\det(\Sigma) = 1 + 2vRa - a^2 - v^2 - R^2 \;\ge\; 0 \tag{6}$$

$$\Longrightarrow\; \underbrace{1}_{\alpha} a^2 \;-\; \underbrace{2vR}_{\beta} a \;+\; \underbrace{(R^2 + v^2 - 1)}_{\tau} \;\le\; 0. \tag{7}$$

Applying the quadratic formula to Equation (7) with coefficients α, β, τ yields

$$a \;\in\; \big[\, vR - \delta,\;\; vR + \delta \,\big], \qquad \delta \;:=\; \sqrt{(1 - R^2)(1 - v^2)}. \tag{8}$$

Finally, applying this into the semi-partial correlation:

$$r(X, Y \mid g) = r(X, Y - \hat{Y}_g) \tag{9}$$

$$= \frac{\mathrm{Cov}(X, Y - \hat{Y}_g)}{\sigma(X)\,\sigma(Y - \hat{Y}_g)} \tag{10}$$

$$= \frac{\mathrm{Cov}(X, Y) - \mathrm{Cov}(X, \hat{Y}_g)}{\sigma(Y - \hat{Y}_g)} \qquad (X \text{ has unit variance}) \tag{11}$$

$$= \frac{v - a\,\sigma(\hat{Y}_g)}{\sigma(Y - \hat{Y}_g)}. \tag{12}$$

The last step substitutes $v := \mathrm{Cov}(X, Y)$ and applies $\mathrm{Cov}(X, \hat{Y}_g) = \mathrm{corr}(X, \hat{Y}_g)\,\sigma(X)\,\sigma(\hat{Y}_g) = a \cdot 1 \cdot \sigma(\hat{Y}_g) = a\,\sigma(\hat{Y}_g)$, where the second equality uses $\mathrm{corr}(X, \hat{Y}_g) = a$ (by definition) and $\sigma(X) = 1$ (since we standardized X).

From here, it remains to address $\sigma(\hat{Y}_g)$ and $\sigma(Y - \hat{Y}_g)$. The former follows directly from our definition of $R := \mathrm{corr}(Y, \hat{Y}_g)$, along with the fact that least-squares residuals are orthogonal to their predictor (i.e, $\mathrm{Cov}(Y - \hat{Y}_g,\, \hat{Y}_g) = 0$):

$$R = \mathrm{corr}(Y, \hat{Y}_g) \tag{13}$$

$$= \frac{\mathrm{Cov}(Y, \hat{Y}_g)}{\sigma(Y)\,\sigma(\hat{Y}_g)} \tag{14}$$

$$= \frac{\mathrm{Cov}(\hat{Y}_g,\, \hat{Y}_g) + \mathrm{Cov}(Y - \hat{Y}_g,\, \hat{Y}_g)}{\sigma(Y)\,\sigma(\hat{Y}_g)} \tag{15}$$

$$= \frac{\mathrm{Var}(\hat{Y}_g)}{\sigma(Y)\,\sigma(\hat{Y}_g)} \qquad (\mathrm{Cov}(Y - \hat{Y}_g,\, \hat{Y}_g) = 0 \text{ by least-squares orthogonality}) \tag{16}$$

$$= \frac{\sigma(\hat{Y}_g)}{\sigma(Y)} = \sigma(\hat{Y}_g) \qquad (\mathrm{Var}(\hat{Y}_g) = \sigma^2(\hat{Y}_g) \text{ and } \sigma(Y) = 1). \tag{17}$$

Hence $\sigma(\hat{Y}_g) = R$ and $\sigma^2(\hat{Y}_g) = R^2$. Then, expanding $\sigma(Y - \hat{Y}_g)$ yields:

$$\begin{aligned}
\sigma(Y - \hat{Y}_g) &= \sqrt{\mathrm{Var}(Y - \hat{Y}_g)} && (18)\\
&= \sqrt{\mathrm{Var}(Y) + \mathrm{Var}(\hat{Y}_g) - 2\mathrm{Cov}(Y, \hat{Y}_g)} && (19)\\
&= \sqrt{1 + R^2 - 2 \cdot \mathrm{corr}(Y, \hat{Y}_g) \cdot \sigma(\hat{Y}_g) \cdot \sigma(Y)} \qquad \left(\text{Since } \tfrac{\mathrm{Cov}(Y,\hat{Y}_g)}{\sigma(Y)\sigma(\hat{Y}_g)} = \mathrm{corr}(Y, \hat{Y}_g) =: R\right) && (20)\\
&= \sqrt{1 + R^2 - 2R^2} && (21)\\
&= \sqrt{1 - R^2} && (22)
\end{aligned}$$

Substituting back into (12) and combining with the bounds on a from Equation (8), we obtain

$$\begin{aligned}
r(X, Y|g) &= \frac{v - aR}{\sqrt{1 - R^2}} && (23)\\
&\in \left[\frac{v - (vR + \delta)R}{\sqrt{1 - R^2}}, \ \frac{v - (vR - \delta)R}{\sqrt{1 - R^2}} \right] \qquad \text{(from Eq. (8))} && (24)\\
&= \left[v\sqrt{1 - R^2} - |R|\sqrt{1 - v^2}, \ \ v\sqrt{1 - R^2} + |R|\sqrt{1 - v^2} \right], && (25)
\end{aligned}$$

which gives the bound in Equation (5). □

Restated Remark on Theorem 1. In particular, a perfectly valid test ($v = 1$) has specificity bounded by $\sqrt{1 - R^2}$; a benchmark with R near ± 1 therefore leaves almost no specificity headroom for *any* test, regardless of construction. This is why Arena CW ($R = 0.98$ on $g = (\text{Arena Overall}, \text{MMLU-Pro})$) caps specificity at ≈ 0.20 for a perfect-validity test, while NoveltyBench Utility ($R \approx -0.33$) admits a much wider ceiling (Figure 4, bottom row).

B. Per-Model Test Scores

In Table 2, we report the mean ± SEM scores for the DAT, CDAT, CDAT-N, CDAT-A, and PACE per model. DAT is scored under GloVe across all valid trials at $T \in \{1.0, 1.5, 2.0\}$ (at 120 total trials per model). CDAT, CDAT-N, and CDAT-A are scored under Sentence-BERT `all-mpnet-base-v2`, where CDAT is novelty restricted to the temperatures passing the false discovery rate $\alpha = .001$ appropriateness gate ("—" if no temperature passed), and CDAT-N and CDAT-A are the ungated novelty and appropriateness scores aggregated over all 50 cues at all three temperatures. PACE is scored under FastText across all valid chains (up to 150 chains per model: 50 seeds × 3 chains).

As a sanity check, our per-model mean ± SEM values fall within the score ranges reported in the original papers. PACE across our 54 models spans ∼0.65–0.76, overlapping the ∼0.69–0.83 range reported by Qiu & Hu (2025) across their 30-model set. Gated CDAT novelty across our models spans ∼62–75, within the ∼55–75 band shown in Figure 3 of Nakajima et al. (2026). Our evaluation extends both sets to a larger number of models.

Table 2. Mean ± SEM of DAT, CDAT, CDAT-N, CDAT-A, and PACE per model, grouped by provider. "—" indicates cells where no valid scores were collected (no temperature passed the appropriateness gate) on the CDAT, or responses were otherwise invalid.

Model	DAT	CDAT	CDAT-N	CDAT-A	PACE
OpenAI					
`gpt-3-5-turbo`	78.24±0.78	73.07±0.54	72.66±0.39	132.92±0.63	0.715±0.004
`gpt-4-1`	86.29±0.27	69.23±0.69	70.10±0.42	140.86±0.55	0.744±0.002
`gpt-4-1-mini`	81.60±0.49	66.88±0.88	67.75±0.58	143.78±0.66	0.730±0.003
`gpt-4-1-nano`	81.93±1.33	72.54±1.01	71.28±0.72	137.84±0.94	0.707±0.004
`gpt-4-turbo`	84.84±1.54	66.49±0.90	66.92±0.57	144.02±0.67	0.732±0.003
`gpt-4o`	82.94±1.15	65.14±1.02	66.26±0.69	144.54±0.78	0.729±0.002
`gpt-4o-mini`	78.70±1.34	70.53±0.76	71.52±0.46	138.41±0.69	0.707±0.003
`gpt-5`	89.33±0.21	69.85±0.85	69.77±0.52	141.96±0.62	0.747±0.002
`gpt-5-4`	91.72±0.19	68.28±0.88	68.63±0.49	143.70±0.54	0.727±0.003

(continued on next page)

(continued from previous page)

Model	DAT	CDAT	CDAT-N	CDAT-A	PACE
gpt-5-4-mini	84.06±0.32	65.21±0.83	65.47±0.52	145.88±0.59	0.734±0.002
gpt-5-4-nano	83.77±0.29	63.20±1.11	63.15±0.58	147.61±0.64	0.678±0.004
gpt-5-mini	82.92±0.36	67.90±0.86	68.02±0.49	143.90±0.57	0.741±0.003
gpt-5-nano	80.66±0.32	62.39±1.04	63.11±0.59	147.95±0.63	0.712±0.003
o3	89.46±0.24	70.09±0.78	70.06±0.44	142.25±0.54	0.748±0.004
o3-mini	76.37±0.42	65.32±1.03	65.67±0.61	144.17±0.71	0.715±0.003
o4-mini	84.44±0.36	68.32±0.81	68.77±0.46	142.61±0.58	0.732±0.003
Anthropic					
claude-3-5-haiku	87.36±0.22	66.62±0.76	67.95±0.43	141.52±0.52	0.726±0.002
claude-3-haiku	78.87±0.23	62.85±0.94	62.53±0.58	146.22±0.65	0.697±0.005
claude-haiku-4-5	85.74±0.26	67.61±0.90	67.80±0.51	143.71±0.65	0.667±0.010
claude-opus-4-5	89.26±0.21	69.71±0.75	69.90±0.38	143.03±0.46	0.742±0.003
claude-opus-4-6	89.70±0.97	—	67.22±0.00	148.68±0.00	0.750±0.002
claude-sonnet-4	86.69±0.16	70.04±0.59	69.78±0.37	141.69±0.53	0.739±0.004
claude-sonnet-4-5	86.67±0.17	66.93±0.84	66.43±0.48	146.01±0.56	0.756±0.002
claude-sonnet-4-6	88.97±0.21	—	—	—	0.755±0.002
Google					
gemini-2-0-flash-001	82.32±0.41	70.72±0.68	70.32±0.41	139.44±0.55	0.730±0.003
gemini-2-5-flash	76.68±0.54	68.54±0.85	68.54±0.49	143.39±0.58	0.742±0.002
gemini-2-5-pro	89.69±0.25	71.18±0.59	71.13±0.34	139.02±0.49	0.761±0.002
gemma-2-27b-it	81.87±0.44	70.46±2.10	70.92±0.61	138.29±0.85	0.694±0.008
gemma-2-9b-it	77.89±1.52	74.09±0.62	72.77±0.51	133.71±0.73	0.728±0.003
gemma-3-27b-it	86.49±0.25	71.75±0.59	72.05±0.33	137.58±0.51	0.728±0.005
Meta					
llama-3-1-70b-instruct	84.78±2.07	71.10±1.06	68.19±0.80	141.56±0.98	0.713±0.004
llama-3-1-8b-instruct	79.88±3.14	—	72.99±0.78	134.84±1.09	0.701±0.006
llama-3-2-1b-instruct	81.20±0.25	52.46±5.27	58.92±1.33	146.46±1.67	0.586±0.017
llama-3-2-3b-instruct	84.11±0.14	64.42±2.47	68.31±0.50	139.37±0.67	0.711±0.005
llama-3-3-70b-instruct	82.47±2.04	68.36±1.71	69.81±0.63	139.24±0.81	0.718±0.004
llama-4-maverick	85.28±0.26	67.34±0.63	67.45±0.44	141.98±0.56	0.707±0.005
llama-4-scout	84.48±1.05	66.90±0.84	67.39±0.50	141.42±0.61	0.696±0.005
Mistral					
mistral-7b-instruct-v0-1	81.20±0.01	69.13±0.96	69.16±0.54	135.43±0.56	0.613±0.011
mistral-large-2407	88.15±0.24	70.71±0.56	70.36±0.34	138.80±0.55	0.737±0.002
mistral-large-2411	81.91±0.40	64.87±0.82	64.54±0.49	146.01±0.54	0.722±0.003
mistral-nemo	78.17±3.01	—	71.09±0.95	137.00±1.28	0.709±0.005
mistral-small-24b-instruct-2501	82.39±2.09	—	73.56±0.76	132.24±0.94	0.717±0.003
Qwen					
qwen-2-5-72b-instruct	72.28±2.64	68.89±2.00	68.98±0.82	138.68±0.92	0.703±0.004
qwen3-14b	81.36±1.76	—	73.89±1.20	131.84±1.70	0.606±0.017
qwen3-235b-a22b	84.95±0.45	68.65±0.91	68.63±0.62	142.18±0.78	0.725±0.003
qwen3-32b	85.18±1.57	—	74.54±1.34	135.62±1.72	0.658±0.019
qwen3-8b	83.31±0.35	67.53±0.80	68.53±0.45	141.87±0.60	0.694±0.004
qwq-32b	82.63±0.50	—	—	—	—
DeepSeek					
deepseek-chat	81.12±0.40	68.33±0.75	67.50±0.53	143.11±0.71	0.729±0.003
deepseek-chat-v3-0324	80.14±1.81	68.36±0.90	68.00±0.55	143.13±0.68	0.730±0.004
deepseek-r1	83.80±0.39	68.75±0.72	69.28±0.45	141.38±0.65	0.720±0.003
Cohere					
command-a	82.28±0.33	62.99±1.12	63.22±0.64	147.80±0.66	0.714±0.003
command-r-plus-08-2024	87.69±0.45	69.61±0.83	69.44±0.49	138.70±0.64	0.722±0.003
NVIDIA					
llama-3-1-nemotron-70b-instruct	84.38±2.71	70.27±3.86	69.62±0.76	140.32±0.93	0.638±0.013
Microsoft					
phi-4	74.29±3.24	—	72.71±0.64	134.34±1.03	0.680±0.006

C. Per-Model Benchmark Scores

Table 3 reports each model's score on every external benchmark used in this study. The general capability proxies are Arena Overall (Chatbot Arena Elo) and MMLU-Pro accuracy (Wang et al., 2024), the creative writing benchmarks are Arena CW (Elo), EQ-Bench CW v3 (Elo) (Paech, 2023), and Mazur CW (Mazur, 2025), the divergent thinking (output-diversity) benchmarks are Hivemind diversity (1− intra-model cosine similarity) (Jiang et al., 2025) and NoveltyBench Utility (NovB.; cumulative-utility score) (Zhang et al., 2025), and the scientific ideation benchmark is the LiveIdeaBench (Ruan et al., 2026) *idea score average* across five dimensions: originality, flexibility, feasibility, clarity, and fluency. Cells marked "—" are missing because the corresponding benchmark does not score that model—future work can expand the coverage of these benchmarks to improve the statistical robustness of observed correlations.

Mazur CW snapshot. Mazur CW scores are transcribed from commit `80b7f17` (an absolute 0–10 mean-rubric leaderboard with 50 graded models). The leaderboard has since been regraded with a new ensemble of judges, eventually deprecating the absolute-rating system entirely in favor of a pairwise Thurstone ranking. We retained this snapshot rather than the latest version because it covers the largest set of models intersecting our evaluation pool ($n = 20$ overlap, vs. ≤ 10 for any later snapshot), keeping per-cell sample sizes comparable to the other creative writing benchmarks.

MMLU-Pro source. MMLU-Pro accuracies are taken from the TIGER-Lab MMLU-Pro leaderboard (CSV at `TIGER-Lab/mmlu_pro_leaderboard_submission`), which is methodologically tied to the original MMLU-Pro paper (Wang et al., 2024). Models without a leaderboard entry receive no MMLU-Pro value and are excluded from the specificity computation for that cell, which is why the per-benchmark specificity sample sizes are smaller than the validity sample sizes (Table 1).

Hivemind mean-similarity estimate. Jiang et al. (2025) does not publish a mean intra-model similarity for each model. Instead, Table 6 in their paper reports, for each of 79 models, the percentage p_b of response pairs whose pairwise cosine similarity falls into each of ten bins covering $[0, 1]$ in 0.1-wide steps. We estimate the per-model mean similarity as the bin-midpoint-weighted sum, and report "Hivemind diversity" as $1 -$ intra-sim.[7], where intra-sim is given by:

$$\text{intra-sim} \;:=\; \tfrac{1}{100}\sum_{b=1}^{10} p_b \cdot m_b, \qquad m_b \in \{0.95, 0.85, \ldots, 0.05\}, \tag{26}$$

Table 3. **Per-model benchmark scores.** Columns: Arena Overall (Elo), MMLU-Pro (accuracy), Arena CW (Elo), EQ-Bench CW v3 (Elo), Mazur CW, Hivemind diversity, NoveltyBench Utility, LiveIdeaBench (5-dim Average). "—" indicates the corresponding benchmark does not score that model.

Model	Arena Ovr	MMLU-Pro	Arena CW	EQ-B. CW	Mazur	Hive.	NovB.	LiveIdea
OpenAI								
`gpt-3-5-turbo`	1223	—	1187	519	—	—	—	—
`gpt-4-1`	1413	0.82	1402	1419	—	—	—	—
`gpt-4-1-mini`	1382	—	1349	1231	—	—	—	—
`gpt-4-1-nano`	1321	—	1306	1034	—	—	—	—
`gpt-4-turbo`	1323	0.64	1322	—	—	0.14	—	6.56
`gpt-4o`	1443	0.75	1423	1484	8.18	0.12	3.27	6.69
`gpt-4o-mini`	1317	0.63	1295	950	6.72	0.11	3.11	6.25
`gpt-5`	1433	0.87	1376	1301	8.60	—	—	—
`gpt-5-4`	1466	0.88	1429	2019	—	—	—	—
`gpt-5-4-mini`	1459	—	1417	1726	—	—	—	—
`gpt-5-4-nano`	1402	—	1336	—	—	—	—	—
`gpt-5-mini`	1389	—	1326	1301	8.31	—	—	—
`gpt-5-nano`	1337	—	1250	866	—	—	—	—
`o3`	1431	0.85	1384	—	8.39	—	—	—
`o3-mini`	1347	0.79	1301	—	6.15	0.17	—	6.51

(continued on next page)

[7]This is exact when within-bin similarity values are concentrated at the bin midpoint and approximate (with worst-case error ± 0.05) under uniform within-bin distributions; the rank order of models is, nonetheless, preserved unless within-bin distributions differ substantially across models

(continued from previous page)

Model	Arena Ovr	MMLU-Pro	Arena CW	EQ-B. CW	Mazur	Hive.	NovB.	LiveIdea
o4-mini	1390	—	1338	—	7.50	—	—	—
Anthropic								
claude-3-5-haiku	1323	0.62	1303	1241	7.35	0.11	2.50	6.37
claude-3-5-sonnet	—	0.78	—	—	—	—	2.36	6.92
claude-3-haiku	1260	0.42	1214	848	—	0.13	—	—
claude-3-opus	—	0.68	—	—	—	—	2.67	6.36
claude-haiku-4-5	1408	—	1384	—	—	—	—	—
claude-opus-4-5	1468	0.87	1462	1769	—	—	—	—
claude-opus-4-6	1496	0.89	1467	1965	—	—	—	—
claude-sonnet-4	1389	0.84	1384	1516	8.09	—	—	—
claude-sonnet-4-5	1451	0.87	1450	1777	—	—	—	—
claude-sonnet-4-6	1462	0.87	1444	1991	—	—	—	—
Google								
gemini-1-5-pro	—	0.70	—	—	—	—	2.73	6.85
gemini-2-0-flash-001	1360	0.78	1346	1252	7.15	0.15	3.17	7.07
gemini-2-0-flash-lite-001	1353	0.72	1345	—	—	—	3.20	6.60
gemini-2-0-pro	—	0.79	—	—	—	—	2.64	7.03
gemini-2-5-flash	1411	—	1399	1255	7.65	—	—	—
gemini-2-5-pro	1448	0.86	1448	1415	8.38	—	—	—
gemma-2-27b-it	1288	0.57	1291	—	—	0.16	3.77	6.65
gemma-2-2b-it	—	0.16	—	—	—	—	4.63	—
gemma-2-9b-it	1265	0.52	1257	920	—	0.17	3.93	—
gemma-3-27b-it	1365	0.68	1348	1256	7.99	—	—	—
Meta								
llama-3-1-405b-instruct	—	0.73	—	—	—	—	3.39	6.62
llama-3-1-70b-instruct	1293	0.63	1257	851	—	0.18	—	6.62
llama-3-1-8b-instruct	1211	0.44	1178	840	—	0.19	3.76	—
llama-3-2-1b-instruct	—	0.12	—	—	—	—	2.81	—
llama-3-2-3b-instruct	1166	0.22	1144	—	—	0.20	3.24	—
llama-3-3-70b-instruct	1318	0.66	1286	—	—	0.13	2.87	6.06
llama-4-maverick	1327	0.81	1307	944	6.20	—	—	—
llama-4-scout	1322	0.74	1290	899	—	—	—	—
Mistral								
mistral-7b-instruct-v0-1	1148	0.26	1104	—	—	0.19	—	—
mistral-large-2407	1313	0.66	1287	1400	6.90	—	—	—
mistral-large-2411	1305	0.68	1276	1082	6.90	0.14	—	6.79
mistral-nemo	1108	0.45	1090	966	—	0.20	—	—
mistral-small-24b-instruct-2501	1273	0.66	1227	—	—	0.18	—	6.41
Qwen								
qwen-2-5-72b-instruct	1302	0.72	1254	—	—	0.14	—	6.62
qwen3-14b	—	—	—	—	—	0.13	—	—
qwen3-235b-a22b	1374	0.83	1324	1379	8.30	—	—	—
qwen3-32b	1347	—	1306	—	—	0.15	—	—
qwen3-8b	—	—	—	—	—	0.12	—	—
qwq-32b	1336	0.69	1296	1262	8.02	—	—	7.06
DeepSeek								
deepseek-chat	1358	0.76	1349	—	—	0.14	—	6.72
deepseek-chat-v3-0324	1395	0.81	1390	1473	7.70	0.14	—	—
deepseek-r1	1398	0.84	1374	1500	8.30	—	—	7.18
Cohere								
command-a	1353	—	1337	1184	—	—	—	—
command-r-08-2024	1249	—	1209	—	—	—	2.98	—
command-r-plus-08-2024	1276	—	1263	—	—	0.23	3.08	—
command-r7b-12-2024	—	—	—	—	—	—	3.35	—
NVIDIA								
llama-3-1-nemotron-70b-instruct	1298	0.63	1277	—	—	—	—	—

(continued on next page)

(continued from previous page)

Model	Arena Ovr	MMLU-Pro	Arena CW	EQ-B. CW	Mazur	Hive.	NovB.	LiveIdea
Microsoft								
`phi-4`	1255	0.70	1210	—	6.26	0.15	—	6.72

D. Greedy Algorithm for the DAT

The greedy algorithm that can trivially "solve" the DAT is given in Algorithm 1. Starting with a random word from a valid vocabulary set V (e.g., the $\sim$42,000 single-token English nouns in WordNet $\cap$ GloVe 840B), the algorithm then proceeds to *minimize the mean cosine similarity* (equivalently, *maximize the mean cosine dissimilarity*, matching Equation (1)) of all subsequent words, while avoiding any repetition.

Figure 5 reports the result of running the algorithm for 120 independent seeds, and scoring each sequence according to Equation (1) under GloVe, FastText, and Sentence-BERT. The three scores are averaged per trial and across embeddings to produce the values in Figure 5. As shown, the algorithm trivially exceeds the distribution of LLM and human scores.

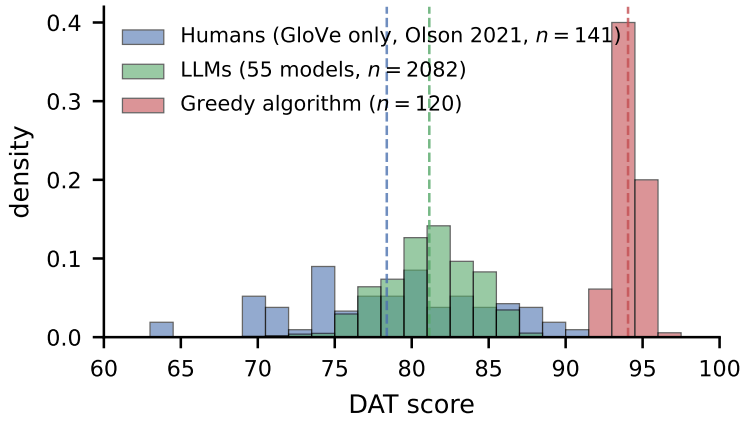

Figure 5. **A simple algorithm outperforms humans and LLMs on the DAT.** Three distributions are reported in this plot: the DAT score distributions for humans (Olson et al. (2021), Study 1A, $n = 141$, GloVe-scored; mean $= 78.4$, std. dev. $= 6.4$), the distribution over our 54-model LLM pool ($n = 2078$ trials at $T = 1.0$; mean $= 83.75$, std. dev. $= 5.13$), and the results of our greedy algorithm over GloVe noun embeddings ($n = 120$; mean $= 94.1$, std. dev. $= 0.9$). Humans are on the GloVe scale only because Olson et al. (2021) only reports final scores—not raw word lists. The dashed lines mark group means.

Algorithm 1 Greedy maximization algorithm for the DAT

Require: vocabulary V, embedding E, number of words n (= 10 for the DAT)
Ensure: word list $W = (w_1, \ldots, w_n)$

$w_1 \sim \mathrm{Uniform}(V)$ // random first word
$W \leftarrow [w_1]$
for $i = 2 \ldots n$ **do**
  $w_i \leftarrow \arg\min_{v \in V \setminus W} \dfrac{1}{|W|} \sum_{u \in W} \cos\big(E(v),\, E(u)\big)$
  $W \leftarrow W \cup \{w_i\}$
end for
return W

E. Prompts

Below, we report the exact prompts used to administer the DAT, CDAT, and PACE. On CDAT, the cue word varies across 50 cues drawn from semantically diverse categories (Physical world, Animals, Food & drink, Body, Emotions, . . .); for PACE

the seed word varies across 50 seeds from the same category set. We show one instantiation with cue/seed `"rock"` for illustration.

E.1. DAT

```
Please enter 10 words that are as different from each other as possible, in all
meanings and uses of the words.  Only use single nouns.  Do not use proper nouns
(names, places, brands).  Do not use variations of the same word (e.g., don't use both
'run' and 'running').

Respond with ONLY a JSON array of exactly 10 words, like:  ["word1", "word2", "word3",
"word4", "word5", "word6", "word7", "word8", "word9", "word10"]
```

E.2. CDAT (cue: rock)

```
Please enter 10 words that are as different from each other as possible, in all
meanings and uses of the words, yet semantically associated with the following cue
word:  "rock".  Only use single nouns.  Do not use proper nouns.  Do not use the
cue word itself or variations of it.  Respond with ONLY a JSON array of exactly 10
words, like:  ["word1", "word2", "word3", "word4", "word5", "word6", "word7", "word8",
"word9", "word10"]
```

E.3. PACE Stage 1 (seed: rock)

```
Starting with the word "rock", generate three different words that directly associate
with this initial word only (not with each other).  Please put down only single words,
and do not use proper nouns (such as names, brands, etc.).  For each word, provide a
brief explanation of its connection to "rock".  Return in JSON format:

{"results":  [{"word":  "", "reason":  ""}, {"word":  "", "reason":  ""}, {"word":  "",
"reason":  ""}]}
```

E.4. PACE Stage 2 (seed: rock, first-association: stone)

```
Starting with the word pair "rock" -> "stone", generate a chain of 20 words where each
new word should be associated with ONLY the word immediately before it.  Generate the
third word based on "stone", then generate the fourth word based on your third word,
and so on.  Please put down only single words, and do not use proper nouns (such as
names, brands, etc.).  For each word, provide a brief explanation of its connection to
the previous word.  Return in JSON format with exactly 20 entries:

{"results":  [{"word":  "stone", "reason":  "<stage-1 reason>"}, {"word":  "",
"reason":  ""}, ... ]}
```

F. Disclosure of Large Language Model Usage

In this paper, large language models (LLMs) were used to assist in the code implementation, plotting and figure generation, and the reporting (but not analysis) of experiment results.